

BUSINESS CASE & ROI ANALYSIS

Data Alchemist

What it costs to run, what it's worth, and the assumptions behind every number

Document type: Business Case & ROI Analysis

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Approach: hard counts are verified from source; business-impact figures are explicitly labeled as projections with disclosed assumptions

What's Verified vs. What's Projected

This document deliberately separates two kinds of numbers. The first kind is **counted directly from the source repository** — module counts, endpoint counts, table counts — and is not in dispute. The second kind is a **projected business-impact estimate**, built on stated assumptions about a typical mid-size data team, and should be read as a reasoned estimate, not a measured result from a paying customer.

Verified (hard counts, not estimates)

11 Checks per profiling run	5 Live DB connectors	94 REST endpoints	\$0 Model training cost
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Source: direct counts from the codebase at time of writing (see Endpoint & Module Inventory workbook for full detail).

Projected (Section 3, assumptions disclosed)

Time-to-detect reduction, manual-effort reduction, and rule-authoring speedup are modeled against a stated baseline for a mid-size data team — not measured from a production deployment, since this is a competition build rather than a live customer engagement.

Why disclose this so explicitly: a business case that hides which numbers are measured and which are estimated is not trustworthy even when the estimates are reasonable. We'd rather be checked than believed.

SECTION 3

Baseline vs. Projected Business Impact

Metric	Baseline (typical, unassisted)	With Data Alchemist	Delta
Time to detect a DQ issue	Discovered downstream, often days later, by a stakeholder noticing a wrong number	Flagged at the next profiling run — minutes, not days	Order-of-magnitude faster
Manual profiling effort per table	~2–3 hours of hand-written SQL per table, per engineer	Automated agent run, engineer reviews the output	~60–70% reduction in hands-on-keyboard time
DQ rule authoring	Fully manual, tribal knowledge, inconsistent coverage across tables	AI-recommended from live profiling results, human-approved before activation	3–5x faster to first draft
Data trust visibility	No single score; "is this table trustworthy" is a Slack thread	One trust score per table, one audit trail per decision	New capability — not a speed-up, a capability gap closed

Assumptions behind this table: mid-size warehouse (50–200 tables), profiling agent run time of ~30–90 seconds per table as observed in this build, and DQ rule activation remaining human-gated by design (the AI proposes, a person decides — this is a deliberate governance choice, not a limitation to be optimized away).

Cost Model & Reusability Value

What It Costs to Run

Cost item	Driver	Notes
LLM API usage	Per-call, tracked in ai_usage_log	Only profiling narratives, rule recommendation, NL-to-SQL, and explainability call an LLM — rule execution and scoring never do
Compute	Docker Compose — postgres, backend, frontend	No GPU requirement; runs on standard application infrastructure
Database	PostgreSQL metadata store	36 migrations, all lightweight metadata/results tables — not a data warehouse replacement
Model licensing	Whichever LLM provider is configured	LiteLLM abstraction means cost can be optimized per client — swap to a cheaper or self-hosted model with one environment variable

Because AI usage is isolated to specific, well-defined call sites (never the deterministic execution/scoring path), the cost surface is small and directly observable via the AI usage ledger — not a black-box "AI tax" on every request.

Reusability & Accelerator Value

- **Cross-client reuse:** a new data source is a 5-method connector implementation, not a rebuild — the agent, API, and UI layers are untouched.
- **Cross-team reuse:** the same platform instance already isolates five different database platforms by connection_id — proven multi-tenancy, not a roadmap item.
- **Faster follow-on engagements:** profiling, rule authoring, and lineage discovery that would otherwise be bespoke per-client scripting become a docker compose up away.

Bottom Line

Data Alchemist replaces ad hoc, per-table SQL scripts and tribal knowledge with a repeatable agentic pipeline: profile automatically, recommend rules from evidence, explain anomalies in plain English, and keep a human in the loop on every decision that matters. The verified footprint (94 endpoints, 5 connectors, 11 automated checks per table, 36 schema migrations) demonstrates this is a working system, not a slide deck — and the cost model shows the AI surface is small, observable, and swappable rather than an open-ended spend.

The honest framing throughout this document — verified counts kept separate from projected business impact — is itself part of the value proposition: a platform whose central purpose is *trust in data* should not ask for blind trust in its own pitch.